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Radio frequency circuit, communication device, and antenna module

Abstract

A radio frequency circuit includes a power amplifier configured to selectively amplify one of a first radio frequency signal and a second radio frequency signal that have different bandwidths, and when the first radio frequency signal is input to the power amplifier, a first bias signal is applied to the power amplifier, and when the second radio frequency signal is input to the power amplifier, a second bias signal different from the first bias signal is applied to the power amplifier.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation of U.S. application Ser. No. 17/820,652 filed on Aug. 18, 2022, which is a continuation of U.S. application Ser. No. 16/897,481 filed on Jun. 10, 2020, which is based on and claims priority of Japanese Patent Application No. 2019-110218 filed on Jun. 13, 2019. The entire disclosures of the above-identified applications, including the specification, drawings and claims are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(1) The present disclosure relates to a radio frequency circuit, and a communication device and an antenna module each including the radio frequency circuit.

BACKGROUND

(2) U.S. Unexamined Patent Application Publication No. 2018/0034152 discloses a configuration of a transceiver front-end circuit (a radio frequency circuit) that includes a primary (main) circuit and a secondary (diversity) circuit.

(3) If the radio frequency circuit disclosed in U.S. Unexamined Patent Application Publication No. 2018/0034152 is applied to a circuit that simultaneously transfers, for example, a radio frequency signal used in a fourth generation mobile communication system (4G) and a radio frequency signal used in a fifth generation mobile communication system (5G), it is assumed that the primary circuit is used as a transfer circuit for 4G, and the secondary circuit is used as a transfer circuit for 5G. In this case, circuit elements such as an amplifier for 4G are disposed in the primary circuit, and circuit elements such as an amplifier for 5G are disposed in the secondary circuit. Thus, circuit elements are disposed in each of the primary circuit and the secondary circuit, and this increases the number of circuit elements, resulting in an increase in the size of the radio frequency circuit. Further, also when the radio frequency circuit disclosed in U.S. Unexamined Patent Application Publication No. 2018/0034152 is applied to a circuit that simultaneously transfers, for example, radio frequency signals that are used in the same communication system, but have frequency ranges in different communication bands, circuit elements are disposed in each of the primary circuit and the secondary circuit, and thus the number of circuit elements increases, resulting in an increase in the size of the radio frequency circuit.

BRIEF SUMMARY

(4) In view of the above, the present disclosure is to address the above problems, and provides a radio frequency circuit, a communication device, and an antenna module that are miniaturized and each include a smaller number of circuit elements such as an amplifier.

(5) In order to provide such a radio frequency circuit, a radio frequency circuit according to an aspect of the present disclosure includes an amplifier configured to selectively amplify one of a first radio frequency signal and a second radio frequency signal that have different bandwidths. When the first radio frequency signal is input to the amplifier, a first bias signal is applied to the amplifier, and when the second radio frequency signal is input to the amplifier, a second bias signal is applied to the amplifier, the second bias signal being different from the first bias signal.

(6) According to the present disclosure, a radio frequency circuit, a communication device, and an antenna module that are miniaturized and each include a smaller number of circuit elements such as an amplifier can be provided.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) These and other advantages and features will become apparent from the following description thereof taken in conjunction with the accompanying Drawings, by way of non-limiting examples of embodiments disclosed herein.
- (2) FIG. 1 illustrates a circuit configuration of a radio frequency circuit and a communication device according to Embodiment 1.
- (3) FIG. 2 is a waveform diagram illustrating a relation of input signal power, an envelope tracking (ET) power supply voltage, and a bias voltage.
- (4) FIG. 3 is a circuit diagram illustrating an example of a circuit configuration of the power amplifier according to Embodiment 1.
- (5) FIG. 4 illustrates a circuit configuration of a radio frequency circuit and a communication device according to Embodiment 2.
- (6) FIG. 5 illustrates a circuit configuration of a radio frequency circuit according to Variation 1 of Embodiment 2, and a peripheral circuit thereof.
- (7) FIG. 6 illustrates a circuit configuration of a radio frequency circuit according to Variation 2 of Embodiment 2, and a peripheral circuit thereof.

DETAILED DESCRIPTION

(8) The following describes in detail embodiments of the present disclosure, with reference to the drawings. Note that the embodiments described below each show a general or specific example. The numerical values, shapes, materials, elements, the arrangement and connection of the elements, and others indicated in the following embodiments are mere examples, and therefore are not intended to limit the present disclosure. Thus, among the elements in the following embodiments, elements not recited in any independent claim are described as arbitrary elements. In addition, the sizes of elements and the ratios of the sizes illustrated in the drawings are not necessarily accurate.

Embodiment 1

- (9) FIG. 1 illustrates an example of a circuit configuration of radio frequency circuit 1 and communication device 5 according to Embodiment 1. As illustrated in FIG. 1, communication device 5 includes radio frequency circuit 1, antenna 2, radio frequency (RF) signal processing circuit (RF integrated circuit (IC)) 3, and baseband signal processing circuit (BBIC) 4.
- (10) Radio frequency circuit 1 includes antenna common terminal 100, power amplifier 10, filter 11, switches 14 and 15, Long Term Evolution (LTE) envelope tracking (ET) power supply 17, and New Radio (NR) ET power supply 18. Radio frequency circuit 1 is a transmitter circuit for transmitting a radio frequency signal generated by RFIC 3 through antenna 2.
- (11) Power amplifier 10 is an example of an amplifier, and can selectively amplify one of a first radio frequency signal (denoted by Sig1 and having bandwidth BW1) and a second radio frequency signal (denoted by Sig2 and having bandwidth BW2) that have different bandwidths. A bias signal is applied to power amplifier 10 to adjust amplification parameters such as a gain.
- (12) For example, the first radio frequency signal is a signal that is used in the fourth generation mobile communication system (4G), and is of a predetermined channel in communication band A used for Evolved Universal Terrestrial Radio Access (E-UTRA). Note that in this specification and the drawings, E-UTRA may be referred to as LTE. Note that communication band A above is Band 41 (a frequency band from 2496 MHz to 2690 MHz) for E-UTRA, for example.
- (13) For example, the second radio frequency signal is a signal that is used in the fifth generation mobile communication system (5G), and is of a predetermined channel in communication band B used for NR. Note that communication band B above is n41 (a frequency band from 2496 MHz to 2690 MHz) for NR, for example.
- (14) Specifically, the bandwidth of the first radio frequency signal used in 4G is a frequency range

of the predetermined channel in the communication band used for E-UTRA, and the bandwidth of the second radio frequency signal used in 5G is a frequency range of the predetermined channel in a communication band used for NR.

(15) Note that signals used in 4G and 5G include signals having frequency ranges not only in licensed bands, but also in unlicensed bands. Examples of the unlicensed bands include LTE-Licensed-Assisted Access (LAA), and NR-U. NR-U is 5G NR that uses 5 GHz or higher, which is defined by the Third Generation Partnership Project (3GPP), and corresponds to the Unlicensed National Information Infrastructure (U-NII) communication band within an unlicensed band defined by Federal Communications Commission (FCC).

(16) 5G signals may be transmitted in the extremely high frequency band (for example, n257). In this case, for example, the first radio frequency signal and the second radio frequency signal may be transmitted in n257 and n261 for 5G, respectively.

(17) Further, one of the first radio frequency signal and the second radio frequency signal may be a wireless local area network (WLAN) IEEE802.11ac signal. In this case, one of LTE ET power supply **17** and ET NR power supply **18** may be a WLAN power supply.

(18) Here, when the first radio frequency signal is input to power amplifier **10**, a first bias signal is applied to power amplifier **10**. On the other hand, when the second radio frequency signal is input to power amplifier **10**, a second bias signal different from the first bias signal is applied to power amplifier **10**.

(19) According to this configuration, a different bias signal is applied to power amplifier **10** based on the bandwidth of an input radio frequency signal, and thus a bias signal is optimized based on the bandwidth of input radio frequency signals. Accordingly, the amplification performance of power amplifier **10** can be increased, and the power consumption thereof can be decreased, as compared with the case where the same bias signal is applied even if the bandwidths of input radio frequency signals are different. Consequently, radio frequency signals in different bandwidths can be amplified by single power amplifier **10**, while the high amplification performance and low power consumption are ensured. Thus, radio frequency circuit **1** can be miniaturized.

(20) LTE ET power supply **17** is an example of a first bias power supply circuit, and outputs a first bias signal for an ET mode (having bias voltage V_{bias1}) to power amplifier **10**, based on control signal S.sub.LTE output by a controller of RFIC **3**.

(21) NR ET power supply **18** is an example of a second bias power supply circuit, and outputs a second bias signal for the ET mode (having bias voltage V_{bias2}) to power amplifier **10**, based on control signal S.sub.NR output by the controller of RFIC **3**.

(22) LTE ET power supply **17** and NR ET power supply **18** are bias signal supply sources for causing power amplifier **10** to operate in an ET amplification mode. The ET amplification mode is for tracking a power amplitude (envelope) of an input radio frequency signal (or an output radio frequency signal) to/from power amplifier **10**, and changing a bias signal (a direct current (DC) bias voltage or a DC bias current) supplied to power amplifier **10** based on the power amplitude.

(23) Note that LTE ET power supply **17** may output a first bias signal for an average power tracking (APT) mode (with bias voltage V_{bias1}) to power amplifier **10**, based on control signal S.sub.LTE output by the controller of RFIC **3**.

(24) Further, NR ET power supply **18** may output a second bias signal for the APT mode (with bias voltage V_{bias2}) to power amplifier **10**, based on control signal S.sub.NR output by the controller of RFIC **3**.

(25) One of LTE ET power supply **17** and NR ET power supply **18** may output a bias signal for the APT mode, and the other of LTE ET power supply **17** and NR ET power supply **18** may output a bias signal for the ET mode. In this case, a ratio of peak power to average power of a signal input to a power amplifier (peak to average power ratio (PAPR) or an average power ratio) tends to be higher in the ET mode than in the APT mode.

(26) Note that the APT mode (an amplification mode in accordance with APT) is for tracking an

average power amplitude of a radio frequency signal calculated for each predetermined period, and changing, in line with the average power amplitude, a bias signal to be supplied to a power amplifier (a DC bias voltage or a DC bias current).

(27) Switch **15** is an example of a first switch, and includes common terminal **15a** and selection terminals **15b** and **15c**. Common terminal **15a** is connected to power amplifier **10**, selection terminal **15b** is connected to LTE ET power supply **17**, and selection terminal **15c** is connected to NR ET power supply **18**. This connection configuration allows switch **15** to switch between application of the first bias signal to power amplifier **10** and application of the second bias signal to power amplifier **10**. Switch **15** is a single pole double throw (SPDT) switch, for example.

(28) Switch **14** is an example of a second switch, and includes common terminal **14a** and selection terminals **14b** and **14c**. Common terminal **14a** is connected to an input terminal of power amplifier **10**, selection terminal **14b** is connected to LTE terminal **31** of RFIC **3**, and selection terminal **14c** is connected to NR terminal **32** of RFIC **3**. According to this connection configuration, switch **14** switches between providing an input of the first radio frequency signal to power amplifier **10** and providing an input of the second radio frequency signal to power amplifier **10**. Switch **14** is an SPDT switch, for example.

(29) Filter **11** includes a first terminal connected to antenna common terminal **100**, and a second terminal connected to an output terminal of power amplifier **10**, and has a passband that includes a frequency range of the first radio frequency signal and a frequency range of the second radio frequency signal.

(30) Antenna **2** is connected to antenna common terminal **100**, and radiates radio frequency signals output from radio frequency circuit **1** through antenna common terminal **100**.

(31) RFIC **3** is an RF signal processing circuit that processes radio frequency signals, and includes LTE terminal **31** and NR terminal **32**. Although not illustrated, RFIC **3** includes a signal processing circuit for LTE, and a signal processing circuit for NR. LTE terminal **31** is an example of a first output terminal, and is connected to the signal processing circuit for LTE. The first radio frequency signal is output through LTE terminal **31**. NR terminal **32** is an example of a second output terminal, and is connected to the signal processing circuit for NR. The second radio frequency signal is output through NR terminal **32**.

(32) Specifically, RFIC **3** processes signals to be transmitted input from BBIC **4** by up-conversion, for instance, and outputs the first radio frequency signal and the second radio frequency signal generated by processing the signals to radio frequency circuit **1**. RFIC **3** outputs control signal S.sub.LTE to LTE ET power supply **17**, when the first radio frequency signal is output through LTE terminal **31**, whereas RFIC **3** outputs control signal S.sub.NR to NR ET power supply **18** when the second radio frequency signal is output through NR terminal **32**.

(33) Note that in the present embodiment, RFIC **3** is an RF signal processing circuit formed into a single chip, which includes the signal processing circuit for LTE and the signal processing circuit for NR. However, in the communication device according to the present disclosure, the signal processing circuit for LTE and the signal processing circuit for NR may be formed into separate chips.

(34) BBIC **4** is a signal processing circuit that processes a radio frequency signal, and is a circuit that processes signals using the intermediate frequency range lower than the frequency range of a radio frequency signal that propagates through radio frequency circuit **1**. The signal processed by BBIC **4** is used as, for example, an image signal for image display, or an audio signal for talk through a loudspeaker.

(35) According to the above configuration of radio frequency circuit **1**, operation of amplifying the first radio frequency signal and operation of amplifying the second radio frequency signal are switched in power amplifier **10** in line with the switching operation of switches **14** and **15**. More specifically, switches **14** and **15** synchronize inputting the first radio frequency signal to power amplifier **10** and applying the first bias signal to power amplifier **10** by switch **14** connecting

common terminal **14a** and selection terminal **14b** and switch **15** connecting common terminal **15a** and selection terminal **15b**. Further, switches **14** and **15** synchronize inputting the second radio frequency signal to power amplifier **10** and applying the second bias signal to power amplifier **10** by switch **14** connecting common terminal **14a** and selection terminal **14c** and switch **15** connecting common terminal **15a** and selection terminal **15c**.

(36) Since switches **14** and **15** are disposed, single power amplifier **10** can amplify the first radio frequency signal and the second radio frequency signal having different bandwidths, while high amplification performance and low power consumption are ensured. Specifically, radio frequency circuit **1** can be miniaturized by adding simple circuits, namely switches.

(37) Note that a divider may be disposed instead of switch **14**, in radio frequency circuit **1** according to Embodiment 1. The divider includes, for example, a common terminal and two divider terminals, divides power of a radio frequency signal input through the common terminal, and outputs the resultant power-divided radio frequency signals through the two divider terminals. Or conversely, the divider outputs, through the common terminal, radio frequency signals input through the two divider terminals. In the above divider, the common terminal is connected to the input terminal of power amplifier **10**, a first divider terminal is connected to LTE terminal **31** of RFIC **3**, and a second divider terminal is connected to NR terminal **32** of RFIC **3**. This connection configuration allows the divider to receive the first radio frequency signal output from RFIC **3** through the first divider terminal, and output the first radio frequency signal to power amplifier **10**. Further, this connection configuration also allows the divider to receive the second radio frequency signal output from RFIC **3** through the second divider terminal, and output the second radio frequency signal to power amplifier **10**.

(38) In this configuration, the operation of amplifying the first radio frequency signal and the operation of amplifying the second radio frequency signal in power amplifier **10** are switched in line with the operation of switching between output of the first radio frequency signal and output of the second radio frequency signal from RFIC **3**.

(39) FIG. **2** is a waveform diagram illustrating a relation of input signal power of a signal input to power amplifier **10**, an ET power supply voltage, and a bias voltage. FIG. **2** illustrates a (carrier) power waveform (input signal power (P_{in}) in FIG. **2**) of a radio frequency signal input to power amplifier **10**, a power supply voltage (ET power supply voltage (V_s) in FIG. **2**) of LTE ET power supply **17** and NR ET power supply **18**, and a voltage waveform (bias voltage (V_{bias}) in FIG. **2**) of a bias signal output from LTE ET power supply **17** and NR ET power supply **18**.

(40) In the radio frequency circuit that includes a power amplifier, power consumption of the power amplifier is a great portion of the power consumption of the radio frequency circuit, and thus an issue to be addressed is to increase efficiency of the power amplifier to reduce power consumption of the radio frequency circuit. The ET method is an example of a technique of increasing efficiency of the power amplifier. ET is a mode of tracking a power amplitude (envelope) of a radio frequency signal, and changing a bias signal (a DC bias voltage or a DC bias current) supplied to the power amplifier in line with the envelope. In particular, a PAPR is increased when a modulation scheme such as orthogonal frequency division multiplexing (OFDM) used for wireless communications is adopted. To amplify and transmit such a modulated signal, a bias voltage is applied to an amplification transistor included in the power amplifier so that the transistor operates in a compression region in response to an input signal at the time of peak power. Specifically, power consumption of the power amplifier can be decreased by changing the bias voltage based on a modulating signal input to the power amplifier to avoid an excessive bias voltage being applied at the time of average power. Note that in the case of ET, the power amplifier is caused to operate in the compression region, and thus the higher the tracking degree of a bias voltage tracking an input modulating signal is, the more likely an output signal is to exhibit severer distortion characteristics.

(41) From the above viewpoint, LTE ET power supply **17** and NR ET power supply **18** each change

an ET power supply voltage (Vs) in line with a power (Pin) amplitude of a radio frequency signal input to power amplifier **10**, as illustrated in FIG. 2. Change in the ET power supply voltage (Vs) changes a bias voltage (Vbias) applied to power amplifier **10** so that the bias voltage tracks a power amplitude of input signal power (Pin) with a predetermined delay time.

(42) Specifically, bias voltage Vbias1 of the first bias signal changes to track a change in the power amplitude of the first radio frequency signal at a first tracking degree, and bias voltage Vbias2 of the second bias signal changes to track a change in the power amplitude of the second radio frequency signal at a second tracking degree. Here, in the present embodiment, the first bias signal and the second bias signal being different means the first tracking degree and the second tracking degree being different, for example.

(43) Table 1 illustrates results of comparing bandwidths, PAPRs, and tracking properties that are to be possessed, for different communication systems (4G and 5G). Table 1 also shows those of WLAN, for reference. The modulating scheme of WLAN is the same as that of 5G (orthogonal frequency division multiple access (OFDMA)), yet PAPR of WLAN may be different from that of 5G.

(44) TABLE-US-00001 TABLE 1 Tracking degree Communication Bandwidth to be possessed system (BW) PAPR (Tracking performance) 4G E-UTRA (LTE) 20 MHz 6 dB Lower than 5G max @SC-FDMA 5G NR 100 MHz 9 dB Lower than 4G max @OFDMA 802.11ac WLAN 160 MHz 9-10 dB Higher than 4G max @OFDMA

(45) As illustrated in Table 1, a bandwidth for 5G is greater than that for 4G, and thus an amplitude change time indicated by the reciprocal of the bandwidth (1/BW) is shorter. A PAPR for 5G is greater than that for 4G. Specifically, if the bandwidth of a radio frequency signal input to power amplifier **10** is different, a PAPR differs. Note that the bandwidths shown in Table 1 are stated in TS 38.101-1, for example.

(46) Accordingly, when the reciprocal of the bandwidth (1/BW) and a PAPR are different, the tracking degree (second tracking degree) of bias voltage Vbias2 tracking an input power amplitude in 5G is to be higher than the tracking degree (first tracking degree) of bias voltage Vbias1 tracking an input power amplitude in 4G.

(47) According to this, when the 5G second radio frequency signals having a relatively wide bandwidth are input to power amplifier **10**, the second bias signal having a higher tracking degree is applied to power amplifier **10**, whereas when the 4G first radio frequency signals having a relatively narrow bandwidth are input to power amplifier **10**, the first bias signal having a lower tracking degree is applied to power amplifier **10**. Accordingly, a bias signal is optimized according to the bandwidth of a radio frequency signal that is input. Thus, power consumption can be reduced as compared with the case where the same first bias signal is applied even if the bandwidths of radio frequency signals that are input are different. Furthermore, distortion characteristics can be improved as compared with the case where the same second bias signal is applied even if the bandwidths of radio frequency signals that are input are different. Consequently, single power amplifier **10** can amplify radio frequency signals having different bandwidths, while high amplification performance and low power consumption are ensured. Furthermore, the amplified first radio frequency signal and the amplified second radio frequency signal pass through one filter **11** and are output through antenna common terminal **100**. Accordingly, the number of amplifiers and the number of filters can be decreased, and thus radio frequency circuit **1** can be miniaturized.

(48) Note that the tracking property (tracking performance) of a bias voltage tracking an input power amplitude is responsiveness of the bias voltage to a change in the power amplitude of a radio frequency signal input to power amplifier **10** (or output from power amplifier **10**), and corresponds to, for example, a transition time of the bias voltage in step response (a rise time or a falling time). Stated differently, a higher tracking degree means quicker responsiveness and a shorter transition time.

(49) Specifically, in radio frequency circuit **1** and communication device **5** according to the present

embodiment in FIG. 1, when the first radio frequency signal having a relatively low PAPR is input to power amplifier **10**, the first bias signal having a relatively low tracking degree is applied from LTE ET power supply **17** to power amplifier **10**. On the other hand, when the second radio frequency signal having a relatively high PAPR is input to power amplifier **10**, the second bias signal having a relatively high tracking degree is applied from NR ET power supply **18** to power amplifier **10**.

(50) Note that in view of the tracking property described above, in radio frequency circuit **1** and communication device **5** according to the present embodiment in FIG. 1, NR ET power supply **18** is desirably disposed closer to RFIC **3** than LTE ET power supply **17**. NR ET power supply **18** is desirably disposed closer to power amplifier **10** than LTE ET power supply **17**. According to this configuration, a control line that connects RFIC **3** and NR ET power supply **18**, and a bias signal line that connects NR ET power supply **18** and power amplifier **10** can be made relatively short, and thus the tracking degree of the second bias signal that is to be higher than the tracking degree of the first bias signal can be improved.

(51) The tracking degree (second tracking degree) of the second bias signal being higher than the tracking degree (first tracking degree) of the first bias signal means that the tracking degree of the second bias signal that NR ET power supply **18** outputs is higher than the tracking degree of the first bias signal that LTE ET power supply **17** outputs. In order to achieve this, for example, a dynamic range of bias voltage V_{bias2} output from NR ET power supply **18** may be greater than the dynamic range of bias voltage V_{bias1} output from LTE ET power supply **17**. For example, the response speed based on a time from when NR ET power supply **18** detects control signal S.sub.LTE until when bias voltage V_{bias2} is output may be lower than the response speed based on a time from when LTE ET power supply **17** detects control signal S.sub.NR until when bias voltage V_{bias1} is output.

(52) Note that in the present embodiment, signals used in 4G and 5G are used as examples of the first radio frequency signal and the second radio frequency signal having different bandwidths, yet the present embodiment is not limited thereto, and the modulation schemes for the first radio frequency signal and the second radio frequency signal may be different. In this case, the first radio frequency signal and the second radio frequency signal may be signals used in the same communication system (for example, both are 4G signals or both are 5G signals). An example is the case where the modulation scheme for the first radio frequency signal is single carrier frequency division multiple access (SC-FDMA), and the modulation scheme for the second radio frequency signal is OFDMA.

(53) The communication system and the modulation scheme may be the same for the first radio frequency signal and the second radio frequency signal having different bandwidths, but the signals may have different PAPRs.

(54) An example is the case where the first radio frequency signal is a signal of a first channel in communication band A for 4G (E-UTRA), and the second radio frequency signal is a signal of a second channel in communication band B for 4G (E-UTRA) having a passband width different from that of communication band A.

(55) Another example is the case where the first radio frequency signal is a signal of a first channel in communication band C for 5G (NR), and the second radio frequency signal is a signal of a second channel in communication band D for 5G (NR) having a passband width different from that of communication band C.

(56) Yet another example is the case where the first radio frequency signal is a signal of the first channel in communication band C for 5G (NR), and the second radio frequency signal is a signal of the second channel also in communication band C. In this case, a signal of the first channel in communication band C for 5G (NR) and a signal of the second channel in communication band C for 5G (NR) may have partially overlapping frequency ranges.

(57) FIG. 3 is a circuit diagram illustrating an example of a circuit configuration of power amplifier

10 according to Embodiment 1. As illustrated in FIG. 3, power amplifier **10** includes, for example, a field effect transistor (FET) that includes a gate (G), a drain (D), and source (S). The gate of the FET is connected to switch **14** (not illustrated in FIG. 3) via input terminal **110**, the drain of the FET is connected to output terminal **120** and switch **15**, and the source of the FET is connected to the ground.

(58) The above configuration allows power amplifier **10** to amplify a radio frequency signal input through input terminal **110**, and output the amplified radio frequency signal to output terminal **120**. Here, when the first radio frequency signal is input to power amplifier **10**, the first bias signal (bias voltage V_{bias1}) is applied from LTE ET power supply **17** to the drain of the FET via switch **15** and a bias input terminal (not illustrated in FIG. 3), whereas when the second radio frequency signal is input to power amplifier **10**, the second bias signal (bias voltage V_{bias2}) is applied from NR ET power supply **18** to the drain of the FET via switch **15** and the bias input terminal (not illustrated in FIG. 3). Note that power amplifier **10** may include input terminal **110**, output terminal **120**, and the bias input terminal.

(59) Note that the first bias signal and the second bias signal that are applied to power amplifier **10** are not limited to so-called bias power supply voltages applied to the drain of the FET. The first bias signal and the second bias signal that are applied to power amplifier **10** may be bias voltages applied to the gate of the FET together with a radio frequency signal. In this case, common terminal **15a** of switch **15** is connected to the gate of the FET. The first bias signal and the second bias signal that are applied to power amplifier **10** may each be both a bias power supply voltage applied to the drain of the FET and a bias voltage applied to the gate of the FET.

(60) Power amplifier **10** may have a configuration in which a plurality of FETs are cascaded. The transistor included in power amplifier **10** may be a bipolar transistor that includes a base, an emitter, and a collector. For example, the transistor included in power amplifier **10** may be a complementary metal oxide semiconductor (CMOS) transistor formed by the silicon on insulator (SOI) process, or may be a transistor made of GaAs or SiGe.

(61) Note that filter **11**, switches **14** and **15**, LTE ET power supply **17**, and NR ET power supply **18** are included in radio frequency circuit **1** according to the present embodiment, but are not necessarily essential to the radio frequency circuit according to the present disclosure.

(62) In communication device **5** according to the present embodiment, antenna **2** and BBIC **4** are not necessarily essential.

(63) The controller that outputs control signals S.sub.LTE and S.sub.NR may not be included in RFIC **3**, may be included in BBIC **4**, or may be included in communication device **5** other than RFIC **3** and BBIC **4**.

(64) In radio frequency circuit **1**, power amplifier **10** and switches **14** and **15** may be formed in a semiconductor integrated circuit (IC). In other words, power amplifier **10** and switches **14** and **15** may be formed on the same IC substrate, and provided as a single chip. The semiconductor IC includes a CMOS, for example, and specifically formed by the SOI process. Accordingly, the semiconductor IC can be manufactured at a low cost. Note that the semiconductor IC may include at least one of GaAs, SiGe, or GaN. Thus, a radio frequency signal having high amplification quality and high noise quality can be output.

(65) Power amplifier **10** may change a frequency range that can be amplified with a predetermined gain or more. Filter **11** may have a passband that can be changed.

Embodiment 2

(66) Embodiment 1 has described an example of a configuration of radio frequency circuit **1** in which separate ET power supplies are disposed to output different bias signals, whereas Embodiment 2 describes an example of a configuration of radio frequency circuit **1A** in which a single ET power supply that can output different bias signals is disposed.

(67) FIG. 4 illustrates a circuit configuration of radio frequency circuit **1A** and communication device **5A** according to Embodiment 2. As illustrated in FIG. 4, communication device **5A** includes

radio frequency circuit **1A**, antenna **2**, RFIC **3A**, and BBIC **4**. Communication device **5A** according to the present embodiment in FIG. **4** is different from communication device **5** according to Embodiment 1 in FIG. **1**, particularly, in the circuit configurations of radio frequency circuit **1A** and RFIC **3A**. In the following, description of the configuration of communication device **5A** according to Embodiment 2 that is the same as that of communication device **5** according to Embodiment 1 is omitted, and a different configuration is mainly described.

(68) RFIC **3A** includes switch **14**. Specifically, RFIC **3A** has incorporated switch **14** into RFIC **3** according to Embodiment 1, and has a single output terminal for outputting radio frequency signals from RFIC **3A** to radio frequency circuit **1A**. In RFIC **3A**, a terminal through which a first radio frequency signal is output is selection terminal **14b**, and a terminal through which a second radio frequency signal is output is selection terminal **14c**.

(69) Radio frequency circuit **1A** includes antenna common terminal **100**, power amplifier **10**, filter **11**, and common ET power supply **19**. Radio frequency circuit **1A** according to the present embodiment is different from radio frequency circuit **1** according to Embodiment 1 in that switches **14** and **15** are not included, and common ET power supply **19** is disposed instead of LTE ET power supply **17** and NR ET power supply **18**. In the following, description of the configuration of radio frequency circuit **1A** according to Embodiment 2 that is the same as that of radio frequency circuit **1** according to Embodiment 1 is omitted, and a different configuration is mainly described.

(70) Power amplifier **10** is an example of an amplifier, and can amplify the first radio frequency signal (denoted by Sig1 and having bandwidth BW1) and the second radio frequency signal (denoted by Sig2 and having bandwidth BW2) that have different bandwidths. A bias signal is applied to power amplifier **10** to adjust amplification parameters such as a gain.

(71) For example, the first radio frequency signal is a signal that is used in 4G, and is of a predetermined channel in the communication band used for E-UTRA (LTE). For example, the second radio frequency signal is a signal that is used in 5G, and is of a predetermined channel in the communication band used for NR.

(72) Here, when the first radio frequency signal is input to power amplifier **10**, a first bias signal is applied to power amplifier **10**. On the other hand, when the second radio frequency signal is input to power amplifier **10**, a second bias signal different from the first bias signal is applied to power amplifier **10**.

(73) Common ET power supply **19** is an example of a bias power supply circuit, and outputs, to power amplifier **10**, the first bias signal (bias voltage Vbias1) and the second bias signal (bias voltage Vbias2) for the ET mode, based on control signal S.sub.LTE/NR that a controller of RFIC **3A** outputs.

(74) When the first radio frequency signal is input to power amplifier **10** through common terminal **14a** in response to switching operation of switch **14** included in RFIC **3A**, the first bias signal is applied from common ET power supply **19** to power amplifier **10**, based on control signal S.sub.LTE/NR (for example, a first digital value) from RFIC **3A**. When the second radio frequency signal is input to power amplifier **10** through common terminal **14a** in response to switching operation of switch **14**, the second bias signal is applied from common ET power supply **19** to power amplifier **10**, based on control signal S.sub.LTE/NR (for example, a second digital value) from RFIC **3A**.

(75) According to the above configurations of radio frequency circuit **1A** and RFIC **3A**, the operation of amplifying the first radio frequency signal by power amplifier **10** and the operation of amplifying the second radio frequency signal by power amplifier **10** are switched based on a radio frequency signal and control signal S.sub.LTE/NR output from RFIC **3A**. More specifically, RFIC **3A** synchronizes the input of the first radio frequency signal to power amplifier **10** and the output of control signal S.sub.LTE/NR (the first digital value), and synchronizes the input of the second radio frequency signal to power amplifier **10** and the output of control signal S.sub.LTE/NR (the second digital value).

(76) According to this, when the 5G second radio frequency signal having a relatively wide bandwidth is input to power amplifier **10**, the second bias signal having a higher tracking degree is applied to power amplifier **10**, whereas when the 4G first radio frequency signal having a relatively narrow bandwidth is input to power amplifier **10**, the first bias signal having a lower tracking degree is applied to power amplifier **10**. Specifically, a bias signal is optimized based on the bandwidth of an input radio frequency signal. Accordingly, power consumption can be reduced as compared with the case where the same first bias signal is applied even if the bandwidths of input radio frequency signals are different. Furthermore, distortion characteristics can be improved as compared with the case where the same second bias signal is applied even if the bandwidths of input radio frequency signals are different. Moreover, the first bias signal and the second bias signal are output from single common ET power supply **19**, and thus the ET power supply can be miniaturized. Accordingly, single power amplifier **10** can amplify radio frequency signals having different bandwidths, while high amplification performance and low power consumption are ensured. Thus, radio frequency circuit **1A** and communication device **5A** can be miniaturized.

(77) Note that the first radio frequency signal and the second radio frequency signal may be output from RFIC **3A** in a time-sharing manner achieved by switch **14**, or may be input to power amplifier **10** being mixed (modulated) without using switch **14**.

(78) When the first radio frequency signal and the second radio frequency signal that are output from RFIC **3A** are input to power amplifier **10** being mixed (modulated), a bias signal in response to control signal S.sub.LTE/NR (for example, a third digital value) output from RFIC **3A** is applied to power amplifier **10**. Here, a bias signal in response to control signal S.sub.LTE/NR (the third digital value) output from RFIC **3A** is a signal obtained by mixing (modulating) a first bias signal component and a second bias signal component. Specifically, power amplifier **10** amplifies an input radio frequency signal obtained by mixing (modulating) the first radio frequency signal and the second radio frequency signal, using a bias signal obtained by mixing (modulating) the first bias signal component and the second bias signal component, so that the input radio frequency signal may be separated into a radio frequency signal obtained by amplifying the first radio frequency signal and a radio frequency signal obtained by amplifying the second radio frequency signal.

(79) Note that RFIC **3A** may not include switch **14**. In this case, similar to RFIC **3** according to Embodiment 1, RFIC **3A** may include LTE terminal **31** through which the first radio frequency signal is output, and NR terminal **32** through which the second radio frequency signal is output.

(80) FIG. 5 illustrates a circuit configuration of radio frequency circuit **1B** according to Variation 1 of Embodiment 2, and a peripheral circuit thereof. As illustrated in FIG. 5, radio frequency circuit **1B** includes antenna common terminal **100**, power amplifier **10**, filters **11**, **12**, and **13**, and common ET power supply **19**. Radio frequency circuit **1B** according to this variation is different from radio frequency circuit **1A** according to Embodiment 2 in that filters **12** and **13** are additionally included. In the following, description of a configuration of radio frequency circuit **1B** according to this variation that is the same as that of radio frequency circuit **1A** according to Embodiment 2 is omitted, and a different configuration is mainly described.

(81) Filter **11** includes a first terminal connected to antenna common terminal **100** and a second terminal connected to the output terminal of power amplifier **10**, and has a passband that includes the frequency range of the first radio frequency signal and the frequency range of the second radio frequency signal. Filter **11** has a passband that includes, for example, Band 3 for 4G (E-UTRA) (transmission band: 1710 MHz to 1785 MHz) and n3 for 5G (NR) (transmission band: 1710 MHz to 1785 MHz).

(82) Filter **12** includes a first terminal connected to antenna common terminal **100**, and has a passband different from the passband of filter **11**. For example, the passband of filter **12** is Band 1 for 4G (E-UTRA) (transmission band: 1920 MHz to 1980 MHz).

(83) Filter **13** includes a first terminal connected to antenna common terminal **100**, and has a passband different from the passbands of filters **11** and **12**. For example, the passband of filter **13** is

Band **40** for 4G (E-UTRA) (band: MHz 2300 to 2400 MHz).

(84) Filters **11**, **12**, and **13** are included in a multiplexer connected to antenna common terminal **100**.

(85) Although not illustrated in FIG. 5, a second terminal of filter **12** and a second terminal of filter **13** may be connected to power amplifiers.

(86) Filters **11** to **13** and antenna common terminal **100** may not be directly connected, and may be connected via switches. The number of filters **11** to **13** is not limited to 3, and is arbitrarily determined.

(87) The above configuration allows radio frequency circuit **1B** not only to simultaneously transmit 4G and 5G signals using power amplifier **10** (EN-DC:E-UTRA-NR Dual Connectivity), but also to aggregate 4G carriers (for example, simultaneously transfer signals having frequency ranges in Band **3**, Band **1**, and Band **40**).

(88) FIG. 6 illustrates a circuit configuration of radio frequency circuit **1C** according to Variation 2 of Embodiment 2, and a peripheral circuit thereof. As illustrated in FIG. 6, radio frequency circuit **1C** includes switch **16**, power amplifier **10**, filter **11**, and common ET power supply **19**. Radio frequency circuit **1C** according to this variation is different from radio frequency circuit **1A** according to Embodiment 2 in that switch **16** is additionally included. In the following, description of a configuration of radio frequency circuit **1C** according to this variation that is the same as that of radio frequency circuit **1A** according to Embodiment 2 is omitted, and a different configuration is mainly described.

(89) Switch **16** includes common terminal **16a** and selection terminals **16b** and **16c**, common terminal **16a** is connected to a first terminal of filter **11**, selection terminal **16b** is connected to antenna **21**, and selection terminal **16c** is connected to antenna **22**. Switch **16** switches between connection of filter **11** to antenna **21** and connection of filter **11** to antenna **22**. Switch **16** is an SPDT switch, for example.

(90) In the above configuration, for example, when common terminal **14a** and selection terminal **14b** are connected, common terminal **16a** and selection terminal **16b** are connected, whereas when common terminal **14a** and selection terminal **14c** are connected, common terminal **16a** and selection terminal **16c** are connected. According to this, when the first radio frequency signal is input to power amplifier **10**, antenna **21** radiates a radio frequency signal output from power amplifier **10**, and when the second radio frequency signal is input to power amplifier **10**, antenna **22** radiates a radio frequency signal output from power amplifier **10**. Thus, isolation of an amplified signal resulting from the first radio frequency signal being amplified by power amplifier **10** from an amplified signal resulting from the second radio frequency signal being amplified by power amplifier **10** can be improved.

(91) As described above, according to Embodiments 1 and 2, radio frequency circuit **1** includes power amplifier **10** configured to selectively amplify one of a first radio frequency signal and a second radio frequency signal that have different bandwidths. When the first radio frequency signal is input to power amplifier **10**, a first bias signal is applied to power amplifier **10**, and when the second radio frequency signal is input to power amplifier **10**, a second bias signal is applied to power amplifier **10**, the second bias signal being different from the first bias signal.

(92) According to this, different bias signals are applied to power amplifier **10** based on the bandwidths of input radio frequency signals, and thus a bias signal can be optimized based on the bandwidth of an input radio frequency signal. Accordingly, amplification performance of power amplifier **10** can be increased and power consumption thereof can be decreased, as compared with the case where the same bias signal is applied even if the bandwidths of input radio frequency signals are different. Consequently, radio frequency signals having different bandwidths can be amplified by single power amplifier **10**, while high amplification performance and low power consumption are ensured. Thus, radio frequency circuit **1** can be miniaturized.

(93) The first bias signal may have a peak to average power ratio (PAPR) that is different from a

PAPR of the second bias signal.

(94) According to this, bias signals having different PAPRs are applied to power amplifier **10** based on the bandwidths of input radio frequency signals, and thus a bias signal can be optimized based on the bandwidth of an input radio frequency signal. Accordingly, radio frequency signals having different bandwidths can be amplified by single power amplifier **10**, while high amplification performance and low power consumption are ensured. Thus, radio frequency circuit **1** can be miniaturized.

(95) A bias voltage of the first bias signal may change to track a change in a power amplitude of the first radio frequency signal at a first tracking degree, and a bias voltage of the second bias signal may change to track a change in a power amplitude of the second radio frequency signal at a second tracking degree, the second tracking degree being different from the first tracking degree.

(96) According to this, bias signals having different tracking degrees to track a power amplitude of a radio frequency signal are applied to power amplifier **10** based on the bandwidth of input radio frequency signals, and thus a bias signal can be optimized based on the bandwidth of an input radio frequency signal. Accordingly, radio frequency signals having different bandwidths can be amplified by single power amplifier **10**, while high amplification performance and low power consumption are ensured. Thus, radio frequency circuit **1** can be miniaturized.

(97) Radio frequency circuit **1** may further include switch **15** configured to switch between application of the first bias signal to power amplifier **10** and application of the second bias signal to power amplifier **10**.

(98) Radio frequency circuit **1** may further include switch **14** configured to switch between input of the first radio frequency signal to power amplifier **10** and input of the second radio frequency signal to power amplifier **10**. Switch **14** and switch **15** may be configured to: synchronize the input of the first radio frequency signal to power amplifier **10** and the application of the first bias signal to power amplifier **10**; and synchronize the input of the second radio frequency signal to power amplifier **10** and the application of the second bias signal to power amplifier **10**.

(99) Single power amplifier **10** can amplify the first radio frequency signal and the second radio frequency signal having different bandwidths, while high amplification performance and low power consumption are ensured. Specifically, radio frequency circuit **1** can be miniaturized by adding simple circuits, namely switches.

(100) A modulation scheme for the first radio frequency signal and a modulation scheme for the second radio frequency signal may be different.

(101) According to this configuration, different bias signals are applied to power amplifier **10** in accordance with the modulating schemes for input radio frequency signals, and thus a bias signal can be optimized in accordance with the modulating scheme for an input radio frequency signal. Accordingly, radio frequency signals modulated using different modulating schemes can be amplified by single power amplifier **10**, while high amplification performance and low power consumption are ensured. Thus, radio frequency circuit **1** can be miniaturized.

(102) The first radio frequency signal may be used in a fourth generation mobile communication system (4G), and the second radio frequency signal may be used in a fifth generation mobile communication system (5G).

(103) Accordingly, 4G-5G EN-DC can be performed using a miniaturized configuration achieved by single power amplifier **10**.

(104) The first radio frequency signal may be used in a fourth generation mobile communication system (4G), and the second radio frequency signal may be used in a wireless local area network (WLAN).

(105) The first radio frequency signal may be used in a fifth generation mobile communication system (5G), and the second radio frequency signal may be used in the 5G.

(106) The first radio frequency signal may be used in a fifth generation mobile communication system (5G), and the second radio frequency signal may be used in a wireless local area network

(WLAN).

(107) The PAPR of the first bias signal may be lower than the PAPR of the second bias signal.

(108) According to this, when the 5G second radio frequency signal having a relatively wide bandwidth is input to power amplifier **10**, the second bias signal having a higher PAPR is applied to power amplifier **10**, whereas when the 4G first radio frequency signal having a relatively narrow bandwidth is input to power amplifier **10**, the first bias signal having a lower PAPR is applied to power amplifier **10**. Accordingly, a bias signal is optimized based on the bandwidth of an input radio frequency signal. Thus, power consumption can be reduced as compared with the case where the same first bias signal is applied even if the bandwidths of input radio frequency signals are different. Furthermore, distortion characteristics can be improved as compared with the case where the same second bias signal is applied even if the bandwidths of input radio frequency signals are different. Accordingly, single power amplifier **10** can amplify radio frequency signals having different bandwidths, while high amplification performance and low power consumption are ensured. Thus, radio frequency circuit **1** can be miniaturized.

(109) According to Embodiment 2, radio frequency circuit **1A** may further include: common ET power supply **19** configured to output the first bias signal and the second bias signal to power amplifier **10**.

(110) Accordingly, the first bias signal and the second bias signal are output from single common ET power supply **19**, and thus the ET power supply can be miniaturized. Consequently, radio frequency circuit **1A** and communication device **5A** can be miniaturized.

(111) According to Embodiment 1, radio frequency circuit **1** may further include LTE ET power supply **17** configured to output the first bias signal to power amplifier **10**; and NR ET power supply **18** configured to output the second bias signal to power amplifier **10**.

(112) Radio frequency circuit **1** may further include filter **11** connected to an output terminal of power amplifier **10**, and having a passband that includes a frequency range of the first radio frequency signal and a frequency range of the second radio frequency signal.

(113) Accordingly, the quality of radio frequency signals to be transmitted which are output from power amplifier **10** can be improved.

(114) Communication device **5** includes RFIC **3** configured to process a radio frequency signal, and radio frequency circuit **1** configured to input the radio frequency signal processed by RFIC **3**.

(115) Accordingly, miniaturized communication device **5** that can amplify a plurality of radio frequency signals having different bandwidths using single power amplifier **10** can be provided.

(116) RFIC **3** may include LTE terminal **31** through which the first radio frequency signal is output to radio frequency circuit **1**, and NR terminal **32** through which the second radio frequency signal is output to radio frequency circuit **1**.

Other Embodiments

(117) The above has described the radio frequency circuits and the communication devices according to the embodiments using the embodiments and variations thereof, yet the radio frequency circuit and the communication device according to the present disclosure are not limited to the above embodiments and the variations thereof. The present disclosure also encompasses other embodiments achieved by combining arbitrary elements in the above embodiments and the variations thereof, variations as a result of applying, to the embodiments and the variations thereof, various modifications that may be conceived by those skilled in the art without departing from the scope of the present disclosure, and also various apparatuses that include the radio frequency circuit and the communication device according to the present disclosure.

(118) Note that the radio frequency circuits and the communication devices according to the above embodiments are typically applied to a system that simultaneously transfers a 4G (E-UTRA) radio frequency signal and a 5G (NR) radio frequency signal, as described above. Examples of a combination of communication band A for E-UTRA and communication band B for NR include (1) Band **41** and n41 stated in Embodiment 1, and (2) Band **20** (transmission band: 832 MHz to 862

MHz, receiving band: 791 MHz to 821 MHz) and n28 (transmission band: 703 MHz to 748 MHz, receiving band: 758 MHz to 803 MHz).

(119) The radio frequency circuits and the communication devices according to the above embodiments are applicable not only to the combinations of communication bands A and B that have at least partially overlapping frequency ranges such as (1) and (2) above, but also to a system that simultaneously transfers radio frequency signals having frequency ranges in two communication bands that do not have overlapping frequency ranges.

(120) Furthermore, the present disclosure also encompasses an antenna module that includes radio frequency circuit 1, and antenna 2 that radiates a radio frequency signal output from radio frequency circuit 1. Accordingly, a miniaturized antenna module that can amplify, with single power amplifier 10, a plurality of radio frequency signals having different bandwidths can be provided. When the radio frequency signals have the extremely high frequency range, an antenna module can be further miniaturized by including antenna 2.

(121) The above embodiments and variations thereof have described, as an example, a configuration adopted when two signals having different bandwidths are transmitted, namely the first radio frequency signal and the second radio frequency signal, yet the configuration of the radio frequency circuit and the communication device according to the present disclosure is also applicable to the configuration adopted when three or more radio frequency signals having different bandwidths are transmitted. Thus, the present disclosure also encompasses a radio frequency circuit and a communication device that are configured to transmit three or more radio frequency signals having different bandwidths, and have the configuration of any of the radio frequency circuits and the communication devices according to the above embodiments and the variations thereof.

(122) Furthermore, for example, in the radio frequency circuits and communication devices according to the above embodiments and the variations thereof, radio frequency circuit elements and lines, for instance, may be interposed between circuit elements and paths that connect signal paths illustrated in the drawings.

(123) The controller according to the present disclosure may be achieved as an IC or a large scale integrated circuit (LSI). Furthermore, a dedicated circuit or a general-purpose processor may perform the method of circuit integration. A field programmable gate array (FPGA) that can be programmed after manufacturing LSIs or a reconfigurable processor that can re-configure the connection and setting of a circuit cell inside an LSI may be used. In addition, if advancement in semiconductor technology or other derivative technology achieves circuit integration technology that replaces LSI, the functional blocks may of course be integrated using such a technology.

(124) Although only some exemplary embodiments of the present disclosure have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the present disclosure.

(125) The present disclosure can be widely used in communication apparatuses such as mobile phones as a multi-band/multi-mode front-end module that adopts EN-DC, in particular.

Claims

1. A radio frequency circuit, comprising: an amplifier configured to amplify radio frequency signals having different bandwidths; and a bias power supply circuit configured to output a bias signal to the amplifier, wherein a bias voltage of the bias signal applied to the amplifier tracks a change in a power amplitude of the radio frequency signal amplified by the amplifier, wherein a degree of tracking the change in the power amplitude corresponds to a bandwidth of the radio frequency signal amplified by the amplifier, wherein the bias power supply circuit is configured to track the

- change in the power amplitude of the radio frequency signal amplified by the amplifier, the degree of tracking by the bias power supply circuit corresponding to the bandwidth of the radio frequency signal amplified by the amplifier, and wherein the bias power supply circuit is configured to adjust the bias voltage of the bias signal in accordance with tracked change.
2. The radio frequency circuit according to claim 1, wherein a peak to average power ratio of the bias signal corresponds to the bandwidth of the radio frequency signal amplified by the amplifier.
 3. The radio frequency circuit according to claim 1, wherein a modulation scheme of the amplifier corresponds to the bandwidth of the radio frequency signal amplified by the amplifier.
 4. The radio frequency circuit according to claim 3, wherein the modulation scheme for a first radio frequency signal amplified by the amplifier is based on a fourth generation mobile communication system (4G), and wherein the modulation scheme for a second radio frequency signal amplified by the amplifier is based on a fifth generation mobile communication system (5G).
 5. The radio frequency circuit according to claim 3, wherein the modulation scheme for a first radio frequency signal amplified by the amplifier is based on a fourth generation mobile communication system (4G), and wherein the modulation scheme for a second radio frequency signal amplified by the amplifier is based on a wireless local area network (WLAN).
 6. The radio frequency circuit according to claim 3, wherein the modulation scheme for a first radio frequency signal amplified by the amplifier is based on a fifth generation mobile communication system (5G), and wherein the modulation scheme for a second radio frequency signal amplified by the amplifier is based on the 5G.
 7. The radio frequency circuit according to claim 3, wherein the modulation scheme for a first radio frequency signal amplified by the amplifier is based on a fifth generation mobile communication system (5G), and wherein the modulation scheme for a second radio frequency signal amplified by the amplifier is based on a wireless local area network (WLAN).
 8. The radio frequency circuit according to claim 1, further comprising: a filter connected to an output terminal of the amplifier, and having a passband that includes a frequency range of a first radio frequency signal amplified by the amplifier and a frequency range of a second radio frequency signal amplified by the amplifier, the first radio frequency signal having a different bandwidth than the second radio frequency signal.
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